

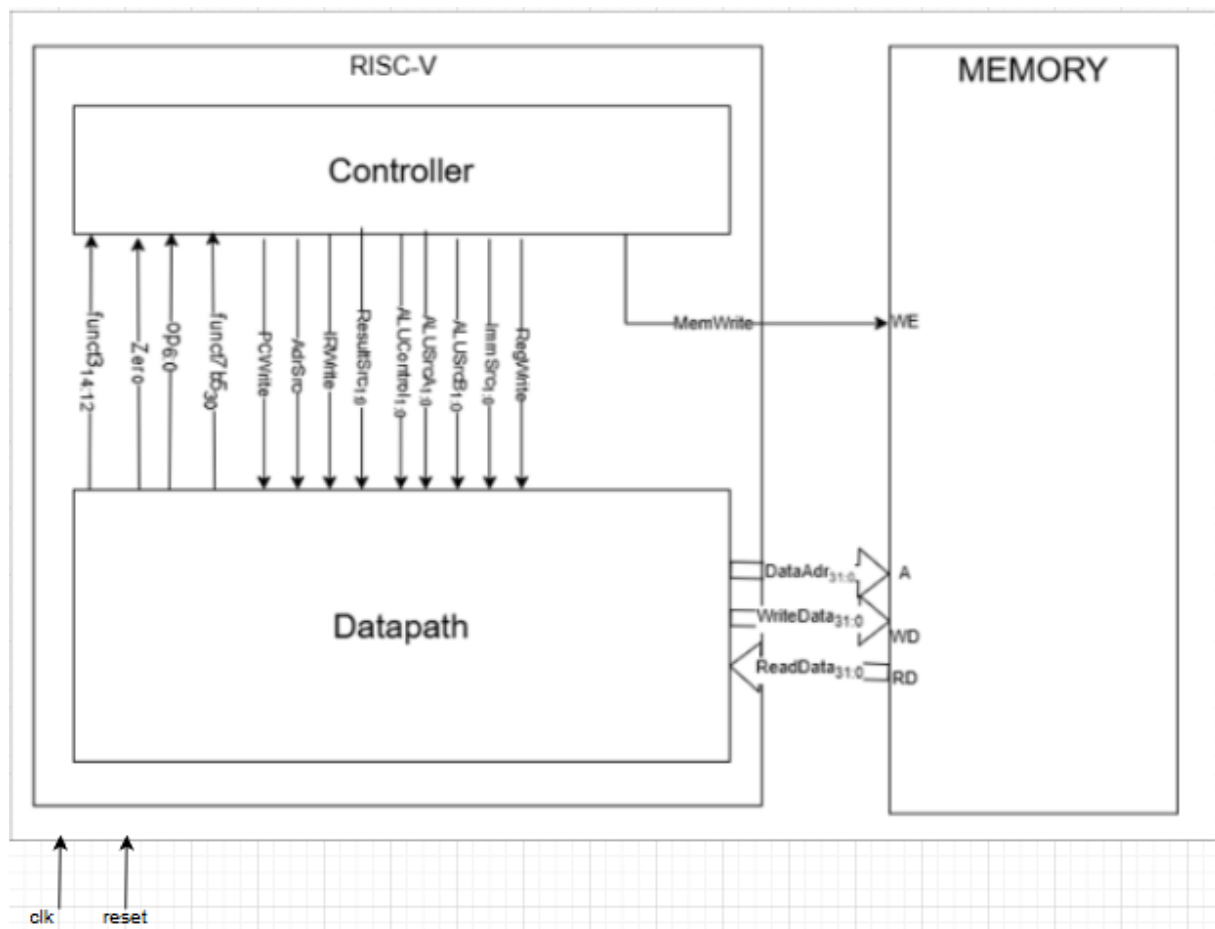
ELE432 - Advanced Digital Design

Preliminary Work 3: Multicycle Processor

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1. I spent 10 hours to complete this preliminary work

2.

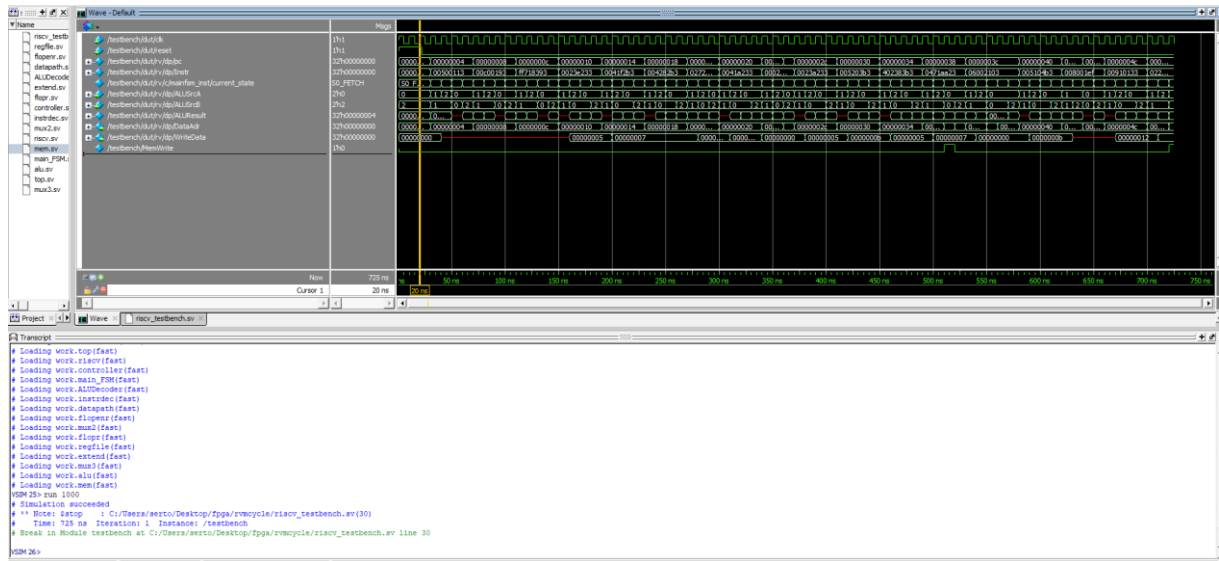


This diagram depicts the high-level structural design of a multi-cycle RISC-V microprocessor. The entire system is housed inside a primary top-level module and comprises two fundamental components:

- **Unified Memory (mem):** A single storage unit holding both the program's code and its associated data. It manages memory access—both reads and writes—directed by the `DataAdr`, `WriteData`, and `MemWrite` lines, and delivers the requested information back through the `ReadData` port.
- **Processor Core (riscv rv):** The central computational hub of the system, which is further separated into two distinct sub-sections:

- Datapath (dp): The execution hardware. This block contains the Arithmetic Logic Unit (ALU), the register file, routing multiplexers, and essential state elements (such as the Program Counter and Instruction register) necessary to carry out mathematical operations and physically move data around.
- Controller (c): The system's command center. It relies on a Finite State Machine (FSM) to interpret incoming instructions and produce the vital control signals required to synchronize and direct the actions of both the memory block and the datapath.

3.



The simulation trace demonstrates that the CPU accurately processes the test program, ultimately storing the data value 25 at memory location 100. This final result verifies the proper functionality of the multi-cycle architecture.

Step (Adım)	PC (Hex)	Instr (Hex)	State (Durum)	Result (Hex)	Result Notes (Açıklama)
7	4	00c00193	S0: Fetch	8	PC + 4 (Yeni PC değeri hesaplanır)
8	8	00c00193	S1: Decode	X (veya Adres)	Branch Target hesaplanır (OldPC + Imm)

Step (Adım)	PC (Hex)	Instr (Hex)	State (Durum)	Result (Hex)	Result Notes (Açıklama)
9	8	00c00193	S8: ExecuteI	0C	ALUResult = x0(0) + 12 = 12 (Hex: C)
10	8	00c00193	S7: ALUWB	0C	Result = ALUOut, bu değer x3 yazmacına yazılır.
11	8	ff718393	S0: Fetch	0C	PC + 4 (Yeni PC değeri hesaplanır)
12	0C	ff718393	S1: Decode	X	Branch Target hesaplanır
13	0C	ff718393	S8: ExecuteI	3	ALUResult = x3(12) + (-9) = 3
14	0C	ff718393	S7: ALUWB	3	Result = ALUOut, bu değer x7 yazmacına yazılır.
15	0C	2,30E+234	S0: Fetch	10	PC + 4 (Yeni PC değeri hesaplanır)
16	10	2,30E+234	S1: Decode	X	Branch Target hesaplanır
17	10	2,30E+234	S6: ExecuteR	7	ALUResult = x7(3) OR x2(5) = 7 (Binary: 011 OR 101 = 111)

Step (Adım)	PC (Hex)	Instr (Hex)	State (Durum)	Result (Hex)	Result Notes (Açıklama)
18	10	2,30E+234	S7: ALUWB	7	Result = ALUOut, bu değer x4 yazmacına yazılır.

5.



I divided waveform into 4 sequential part. From top to bottom the simulation time propagates.

6.

<https://github.com/OmerFarukSert/ELE432-Pre3>